The Indian Monsoon

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Introduction to the Indian Monsoon

The word monsoon comes from the Arabic word "mausim", meaning season. A monsoon is a seasonal reversal of wind direction that brings a distinct rainy season. The Indian monsoon is the most important weather system for the Indian subcontinent. India receives roughly 70 to 75 percent of its annual rainfall during the four months from June to September, and the monsoon is called the real finance minister of India because the farm economy, river flows, reservoirs, electricity generation and prices of food depend on it. The average annual rainfall of the country is about 87 centimetres (long-period average of about 868 millimetres). The India Meteorological Department (IMD), founded in 1875, forecasts the monsoon each year.

What Causes the Monsoon?

The monsoon is caused mainly by the differential heating of land and sea. During summer, the Indian landmass heats up much faster than the surrounding Indian Ocean, Arabian Sea and Bay of Bengal. The hot air over land rises, creating a low-pressure area over north-west India, while the sea remains cooler with higher pressure. Moist winds from the high-pressure ocean rush toward the low-pressure land. Because of the Earth's rotation (the Coriolis effect), the winds blowing from the southern hemisphere across the equator turn to the right and become south-westerly, which is why the summer monsoon is called the southwest monsoon. The moist air is forced to rise over mountains such as the Western Ghats and the Himalayas, cools, condenses and falls as heavy rain, called orographic rainfall. Several other factors influence it: the shift of the Inter-Tropical Convergence Zone (ITCZ) northward over India, the heating of the Tibetan Plateau, the high-altitude jet streams, the ocean-atmosphere pattern El Nino (warming of the Pacific Ocean, which usually weakens the monsoon and causes droughts), La Nina (usually good for the monsoon) and the Indian Ocean Dipole.

The Southwest Monsoon

The southwest monsoon arrives in two branches. The Arabian Sea branch hits the Western Ghats first, bringing heavy rain to the west coast, Kerala, Karnataka, Goa and Maharashtra. The Bay of Bengal branch moves toward north-east India, then turns west along the Ganga plain; the Khasi Hills in Meghalaya receive some of the heaviest rainfall in the world at Mawsynram and Cherrapunji.

When Does the Southwest Monsoon Reach Kerala?

The southwest monsoon normally reaches Kerala around 1 June, and Kerala is therefore called the gateway of the monsoon. The IMD's normal date for onset over Kerala is 1 June, with a standard deviation of about seven days. The arrival is announced when conditions such as widespread rainfall over several stations in Kerala, specific wind patterns and cloud cover are met. The onset in recent years has varied: it arrived on 30 May in 2024, as early as 24 May in 2025, and on 8 June in 2023. From Kerala the monsoon advances northward and normally covers the entire country by about 8 July, and it begins withdrawing from north-west India in mid-September, retreating completely by around mid-October. The sudden heavy beginning of rain is called the burst of the monsoon.

The Northeast Monsoon and Tamil Nadu

After the southwest monsoon withdraws, the winds reverse again and blow from the land toward the sea, from the north-east. This is the northeast monsoon, also called the retreating monsoon, which lasts from October to December. As these winds travel over the warm Bay of Bengal, they pick up moisture and bring rain to the south-eastern coast of India, including Tamil Nadu, coastal Andhra Pradesh, Puducherry and south Karnataka.

Why Does Tamil Nadu Get Most of Its Rain From the Northeast Monsoon?

Tamil Nadu gets most of its rain from the northeast monsoon because of its geography. During the southwest monsoon, the Western Ghats on the west block the moisture-bearing winds, and Tamil Nadu lies in the rain-shadow region on the leeward side, so it receives relatively little rain from June to September. During October to December, the north-east winds blow from the Bay of Bengal onto the east-facing Coromandel coast, carrying plenty of moisture, and the rain-bearing cyclonic depressions and low-pressure systems of the Bay of Bengal also strike the coast at this time. As a result, the northeast monsoon provides nearly half, about 48 percent, of Tamil Nadu's annual rainfall, which is about 920 millimetres. Chennai receives most of its yearly rain in October, November and December. The heavy rains bring floods too: Chennai suffered devastating floods in November and December 2015, and again during Cyclone Michaung in December 2023. The state's reservoirs, tanks (eris) and groundwater are replenished by this season.

Impact on Agriculture and Economy

The monsoon is vital for kharif crops such as rice, maize, cotton, sugarcane and pulses, which are sown in June and July. About half of India's farmland still depends on rain rather than irrigation. A good monsoon raises farm income and rural demand, while a poor or delayed monsoon can cause drought, rising food prices and lower economic growth. Monsoon rain also fills reservoirs for drinking water and hydroelectric power.

Forecasting and Cultural Importance

The IMD issues long-range forecasts in April and updates in May and June, and uses satellites, radars, supercomputers and the Mission Mausam programme, approved in 2024, to improve forecasts. The monsoon inspires literature, music and festivals across India, from classical ragas such as Megh Malhar to poems in Tamil Sangam literature, and festivals like Onam in Kerala and Aadi Perukku in Tamil Nadu celebrate rivers filling with water.

Challenges and Future

Climate change is making monsoon rainfall more erratic, with longer dry spells punctuated by extremely heavy downpours. Better forecasting, water storage, drought-resistant crops and rainwater harvesting are important adaptations.

Quick Facts

- The monsoon is caused by differential heating of land and sea, producing seasonal reversal of winds.
- The southwest monsoon normally reaches Kerala around 1 June.
- Tamil Nadu receives nearly half of its annual rain from the northeast monsoon in October to December.
- The southwest monsoon provides roughly 70 to 75 percent of India's annual rainfall.
- The India Meteorological Department was established in 1875.